ComMarker

B6&B6MOPA USER MANUAL

ComMarker B6&B6 MOPA Laser Engraver

Machine Assembly Guide

Operation Tutorial



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Safety Warning 

Before using the laser engraving machine, please read this safety guide carefully.

■ **Do not leave the device unattended while in operation.** Ensure it is functioning properly at all times.
■ Check the machine for damage every time before you use it. Do not operate it in any way when any damage or defect is found.
■ Inspect the machine for damage before each use. Do not operate if any damage or defects are detected. Ensure the workspace is clean and level.
■ Do not disassemble or alter the machine's structure without authorization. Do not modify or decompile its operating system.
■ Keep the interior of the machine clean. Accumulated residues and debris from cutting and engraving can be hazardous and may cause a fire. Regularly clean the work area to remove chippings and residues.

1 Laser Safety

• The machine uses Class IV lasers, which are extremely powerful and can cause eye injuries or burn the skin. It is recommended to wear laser safety goggles when operating the laser engraver.
•Avoid direct exposure of your skin to Class IV laser beams, especially at close range.
•Teenagers must be supervised by a parent or guardian while using the machine.
•Do not touch the laser engraving beam while it is active.

2 Material Safety

•Do not engrave materials with unknown properties.
•Recommended materials for engraving include:

- Metals: stainless steel, aluminum, bronze, brass, alloys, etc.
- Plastics: PP , PE, ABS, etc.

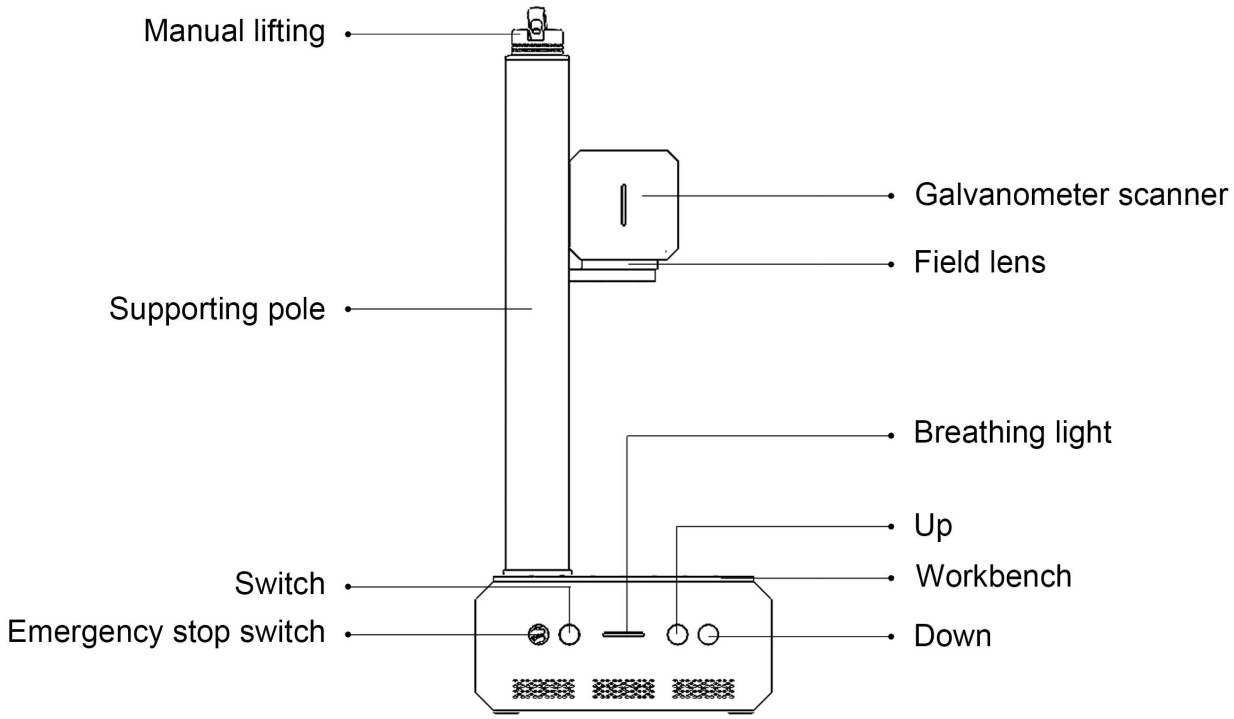
3 Use Safety

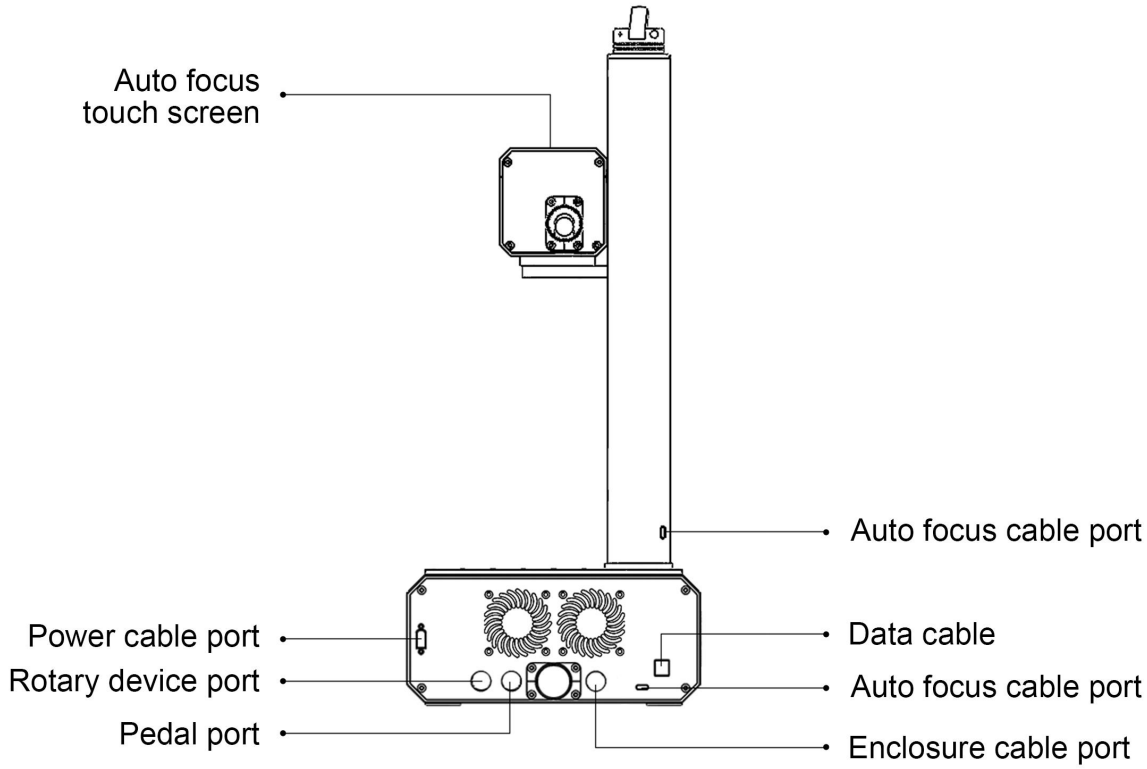
•It is forbidden to point the laser to people, animals or any combustible object, whether it is in working condition or not.
•Use this laser engraving device only in accordance with all applicable local and national laws and regulations.
•Use this device only in accordance with this instruction manual and engraving software manual.
•Any untrained personnel who might be near the device must be informed the danger of the machine before operation

4 Electric Safety

•Only use this device with a compatible and stable power supply with less than 5% fluctuation in its voltage.
•Turn on the power to this device when it is well grounded.
•The area around this laser engraver device should be kept dry, well ventilated, and environmentally controlled to keep the ambient temperature between 40–95°F (5–35°C). The ambient humidity should not exceed 70%.

Product Structure





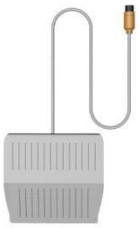
List of items

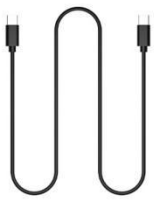

B6 Laser Engraver


Power Adaptor


Data Cable


Rotary Drive


Foot Switch


Type - C


Riser


Allen Wrench


Placing Helper

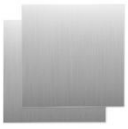

Ruler


Goggles


OTG Adapter*2


USB Drive


Paperboard


Steel For Testing


Screw M5*14 x6


Screw M5*10 x12


Screw M4*12 x4


Screw M5*12 x2

Specifications

Model	20W	30W	50W	20w MOPA	30W MOPA	60W MOPA
Laser Type	Fiber			Mopa fiber		
Working Area	15cm x 15cm（Optional: 11cm x 11cm/15x 15cm/30x 30cm）					
Wavelength	1064nm					
Frequency	20-60kHz	40-60kHz	50-80kHz	1-4000kHz		
Pulse Width	70-110ns	100-120ns	80-110ns	2-500ns		
Max Pulse Energy	0.7mJ	0.75mJ	1mJ	0.8mJ	0.8mJ	2mJ
Cooling Type	Air Cooling					
Voltage:	110~130V or 200~240V					
Dimension(L*W*H)	53 x 43 x 24cm^3					
Weight (KG)	12.5kg			13.5kg		

Assembly of Machine

1. Place the parts on the table



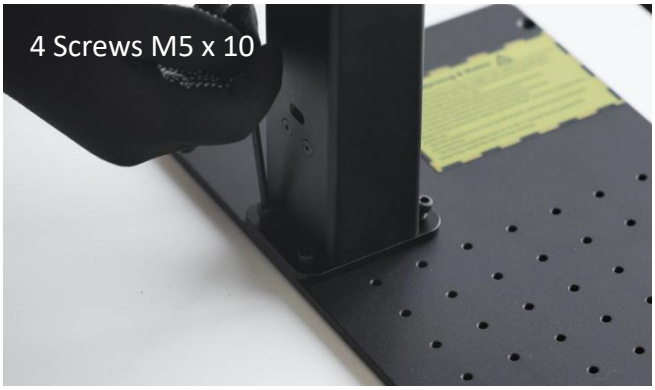
2. Fix the supporting shelf onto the supporting pole



3. Put the laser head on the supporting shelf and fasten the screws to fix the laser head



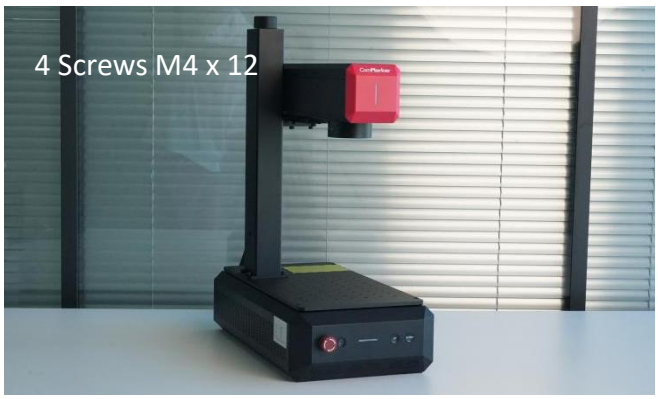
4. Fasten the supporting pole with the screws



5. Put the split machine on the table. The machine can be used at this state.



6. It can also be assembled as an integrated machine with screws



Note:

1) Adjusting the focus is a necessary prerequisite for laser engraving, please press the "Auto" button or manually move the laser head to adjust focus after placing the engraving material

2) Make sure to remove the plastic protective cover of the field lens before engraving to avoid damage to the field lens caused by laser burning.

Assembly of Auxiliary Riser (Optional)

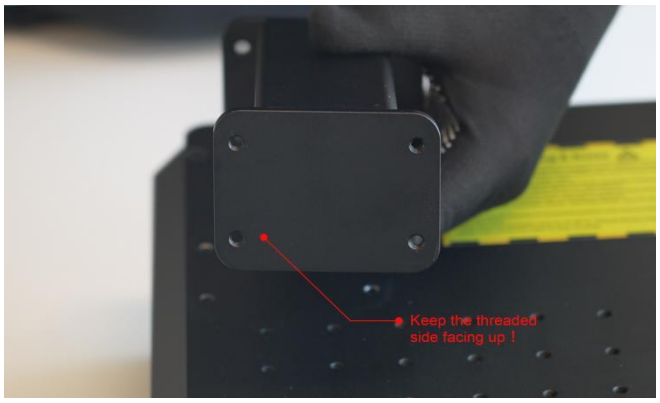
1. In case of engraving a big object, the supporting pole needs to be extended with a riser



2. Disassemble the supporting pole



3. Make sure the side with the threaded holes face upwards.



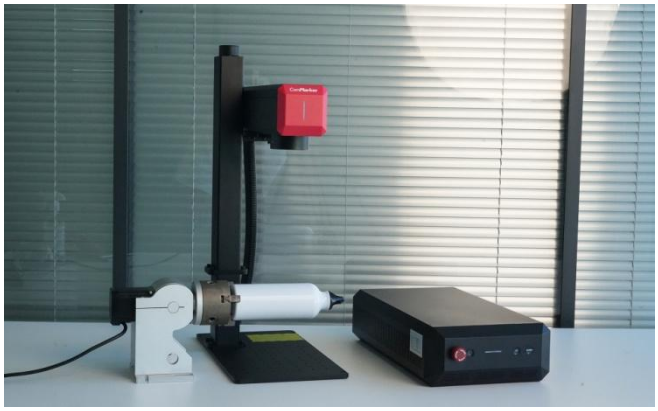
4. Fix the riser on the base board.



5. Connect the riser with the original pole with screws



6. The laser head is raised with enough focus length for bigger objects



Note:

1) It takes about 1 second of pressing the switch button before the machine starts to run.

Replacement of the 300mm Field Lens

The relative position of laser head needs to be adjusted so as to achieve bigger engraving area. The specific operation steps are as follows:

Note: The 300mm field lens is not included as standard and needs to be purchased separately

1. Disassemble field lens



2. Fasten the 300mm screws lens



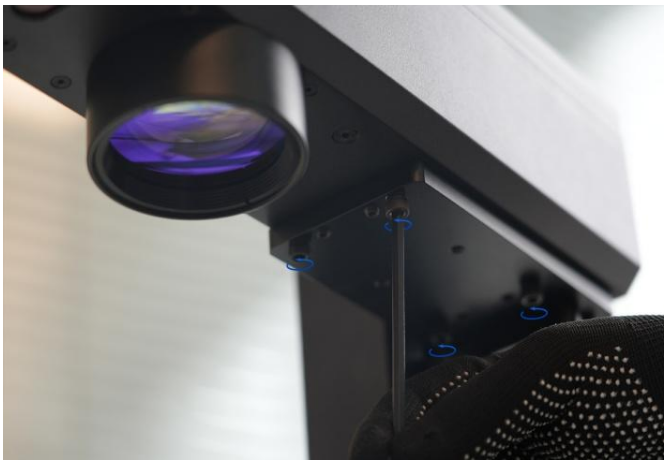
3. Disassemble the laser head by loosening the screws.



4.Move the laser head backward and refix it in the new position.



5.Loosen the four screws on the supporting shelf (note: just loosen them, do not remove them).



6. Adjust the set screws to ensure that the laser head remains level (can use a ruler for measurement)



*Note:**The screws at the front of the supporting plate should not be tightened, and it is not necessary to fasten the two screws at the back of the supporting plate***

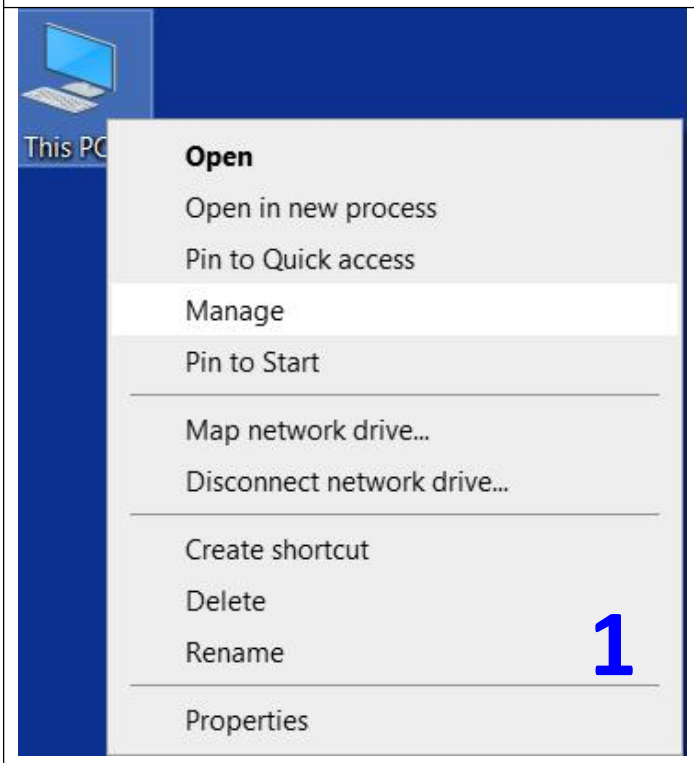
Machine Operation Instructions

1. Install the driver

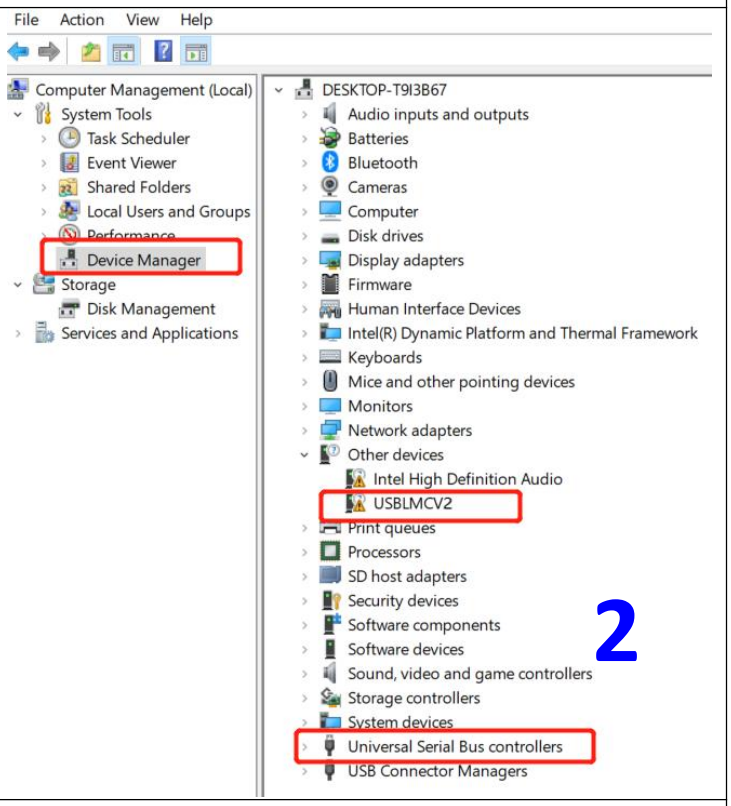
Note: please conduct the following operation before installing the driver

1) The machine is on; 2) Keep the emergency button not pressed; 3) Data cable with the computer is connected ; 4) The Type-C data cable connecting the column to the back of the machine is attached

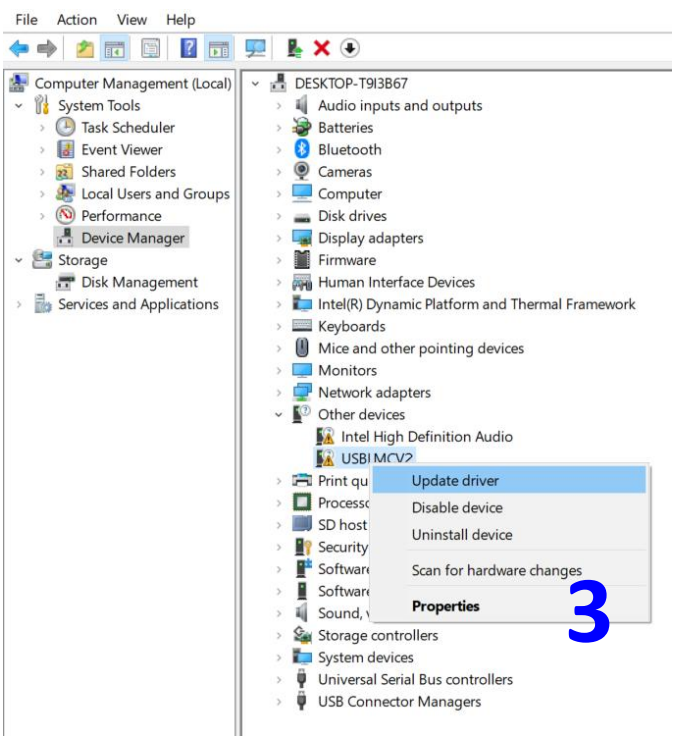
1.Right-click "this PC", click "Manage"



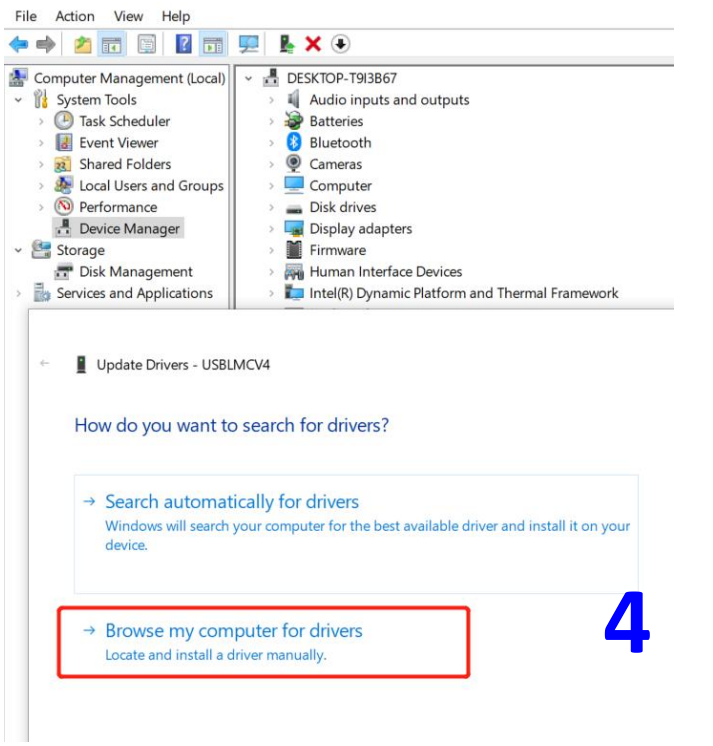
2.Click "Device Manager", find **“USBLMCV2”**, **“USBLMCV4”** or **“Unknown Device”**in **“Other devices”**or **“Universal Serial Bus devices”**



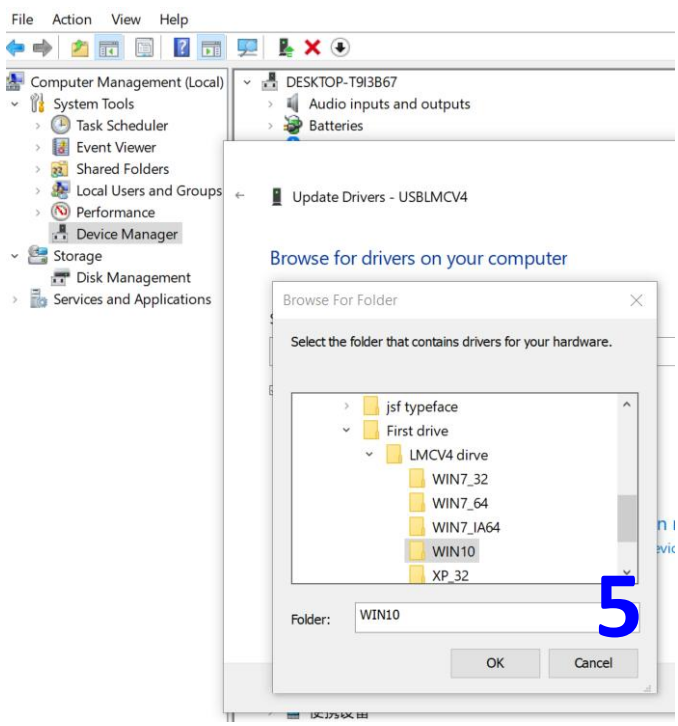
3.Right-click **“USBLMCV2”**, **“USBLMCV4”** or **“Unknown Device”** and "Update driver"



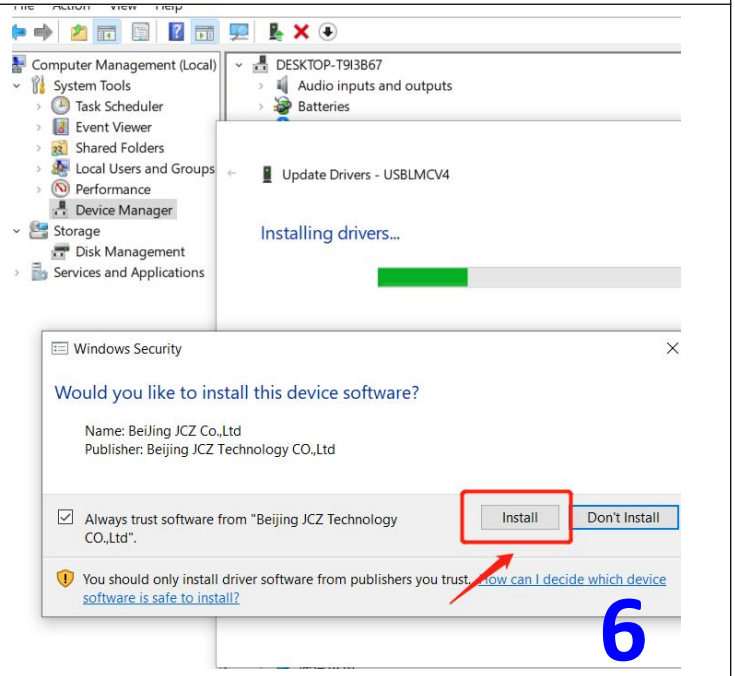
4.Select "Browse My Computer for Drivers"



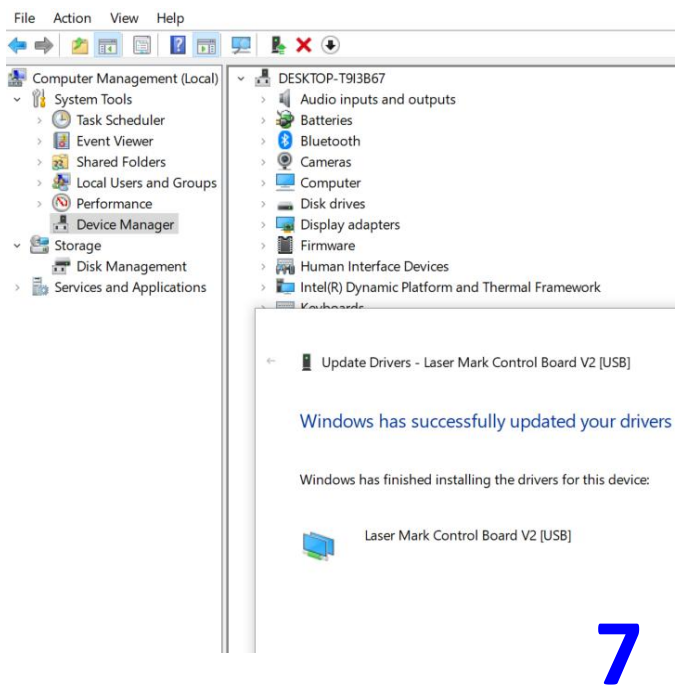
5.Find the corresponding file with window system in the USB drive, click OK



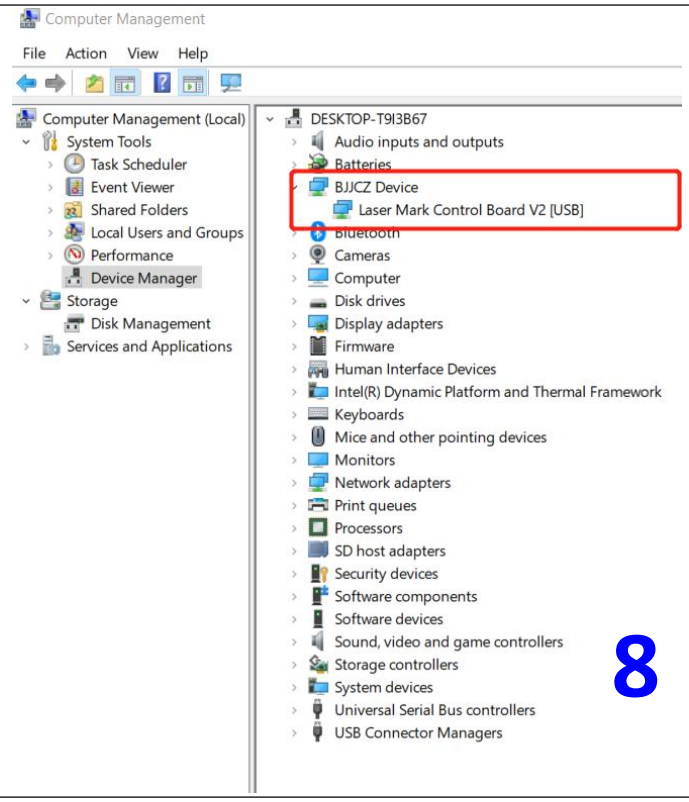
6.Click "Install"



7.Click “finish” if it is successful



8.It will display "BJJCZ Device" in Device Manager

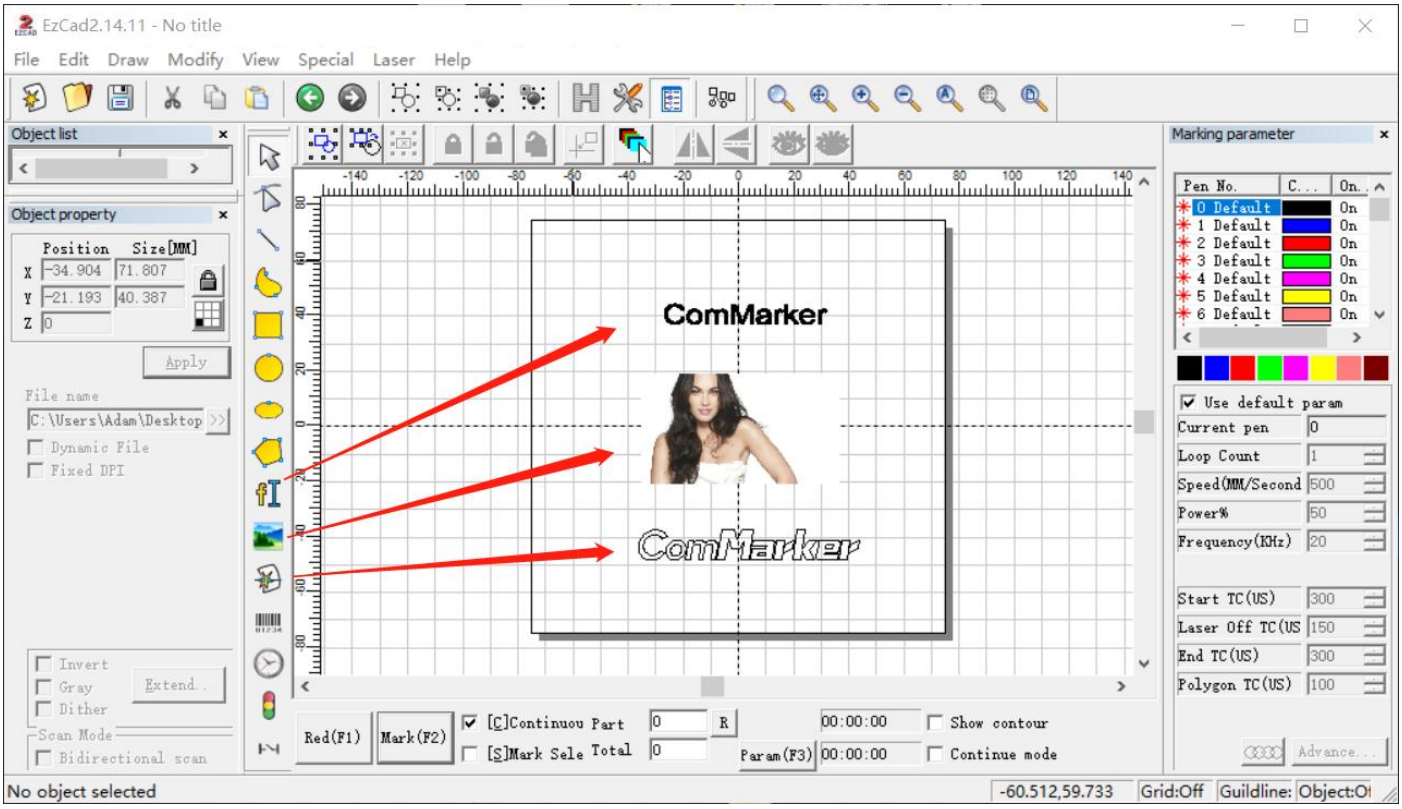


Noted: *It is recommended to copy and paste the files of the USB disk to the computer desktop or other computer disks.

Note: If the driver installation is unsuccessful, please refer to FÃQ

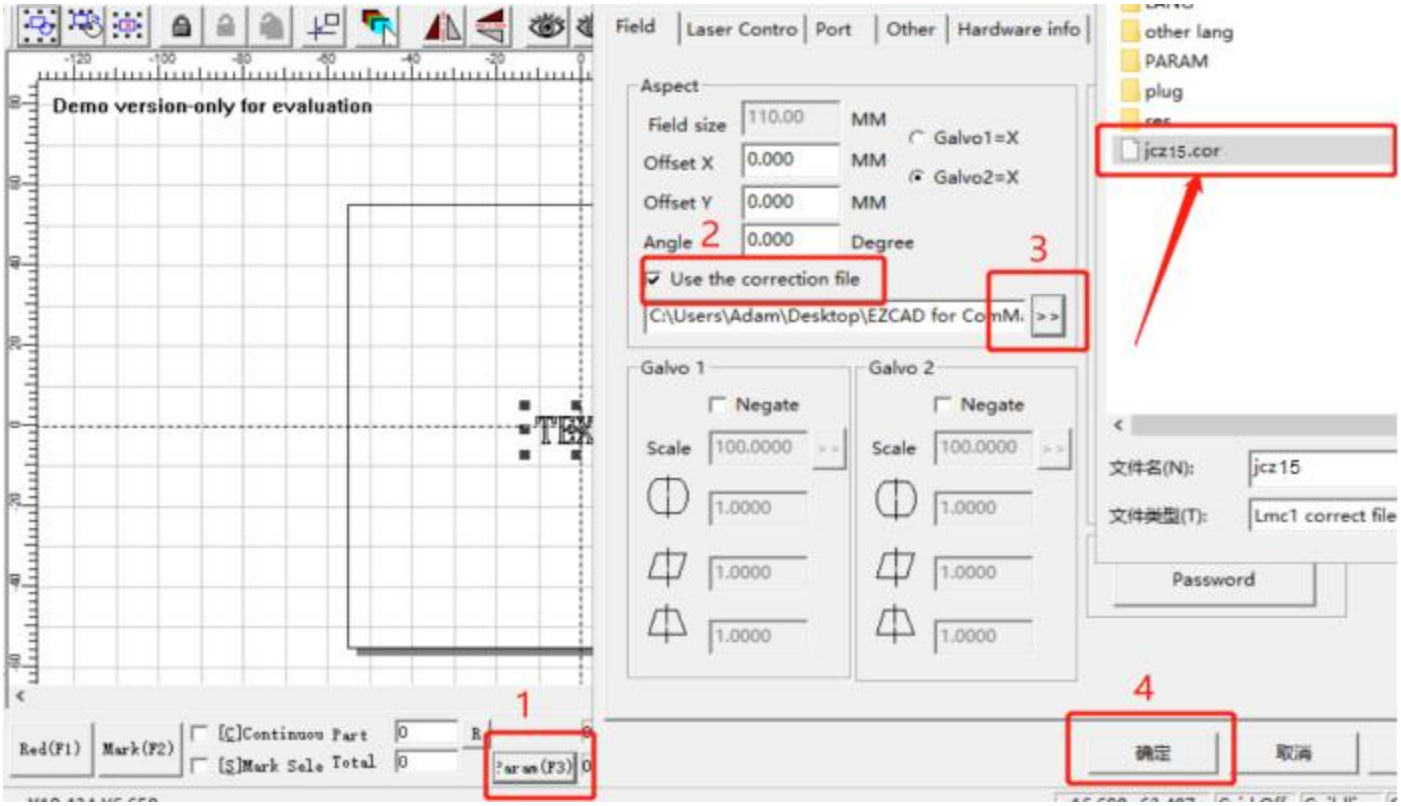
2. Open the software

Open the USB folder, open the "EZCAD for ComMarker" folder, open the software “EZCAD2”. You can enter "text", import "picture" or "vector file" as shown in the figure below. It is recommended to select the text to test first.



3. Import the correction file

Select "Parameter (F3)", tick "Use Correct File", select the file ending in .cor in the USB flash drive (or have been copied to the computer disk), and click "OK (确认)".



4. Adjust the focus

For B6 pro: press the “Auto” button to autofocus.

Note: Please refer to next page “Auto Focus Setting” if it the focus is not right

The sensor is not at the center of the engraving area. Please place the material directly under the red dot before pressing the “Auto” button for accurate focusing. After auto-focusing, the material can be moved if necessary **For B6**

Press up or down button to lift or lower the laser head until the two red light point overlap, this is the best focal point.

The measuring focal length of this machine is:

_____(70*70), _____(110*110), _____(150*150),_____(200*200), _____(300*300)

the measuring distance is the length from the bottom line of the laser head to the surface of the material. This parameter is measured manually, and each device has different focal lengths due to changes in laser sources and field lenses.

Note: Please refer to FAQ if the focal point does not match with the overlapping point



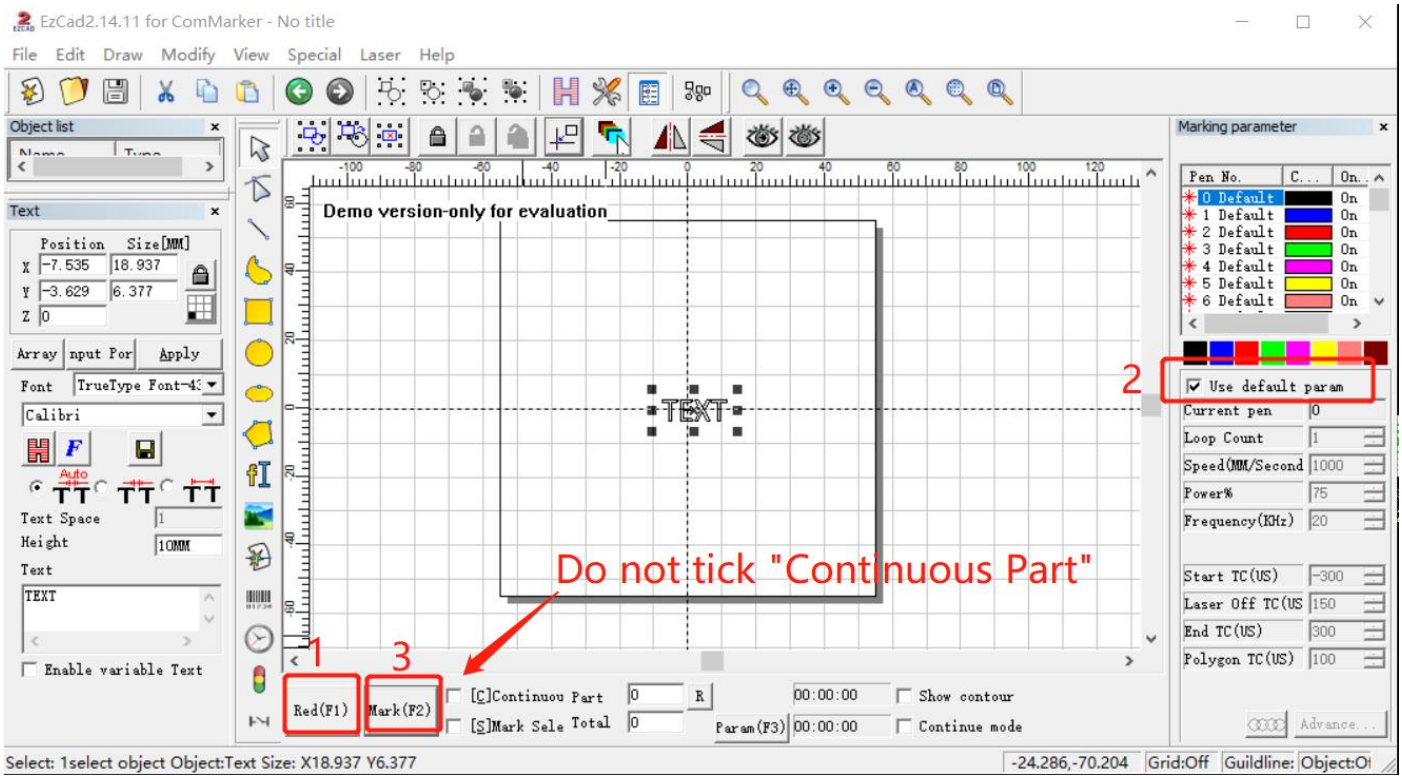
B6 Pro



B6

5. Marking test

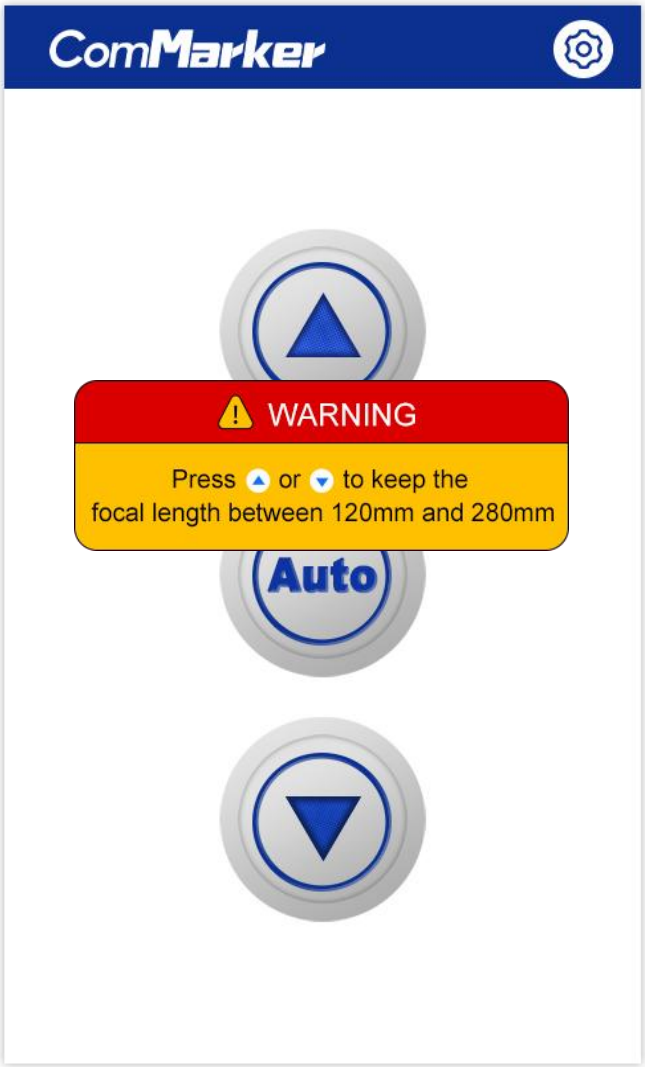
Place the test engraving material on the machine. Click "Red (F1)", the area shown by the red light is the marking area. Tick “Default Parameters". Click "Mark (F2)" to start marking.



Auto Focus Setting(Optional)

Setting of auto focus software is optional because the machine was well set before sending out. The following instruction is for your reference just in case something goes wrong with the machine

Controlling Page



Up: the laser head will go up when pressing the button

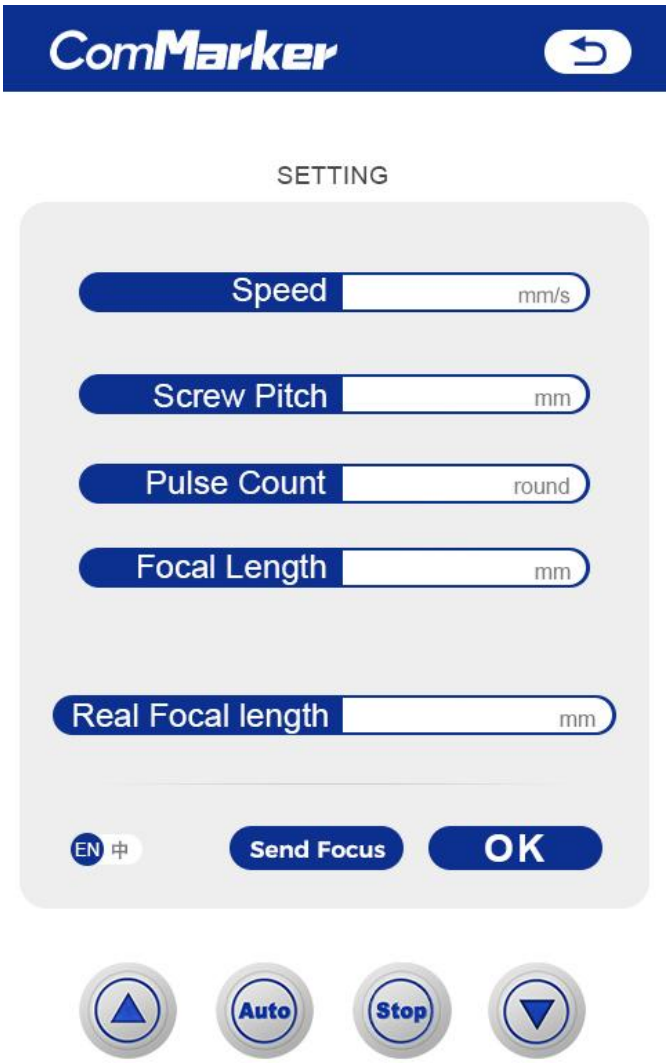
Auto focus: the laser head will move the right focal length according to the original setting

Down: the laser head will go down when pressing the button



WARNING: The sensor in the machine only works in the range of 120mm to 280mm; this warning will pop up while it is out of the range. Please lift up or down manually to adjust the focus distance in between 120mm and 280mm.

Setting Page



Speed: **12.5mm/s.** This is an optimized parameter adjusted by the manufacturer.
Screw pitch: **8mm** for ComMarker B6. Indicate distance between two adjacent threads on a screw.
Pulse count： 200 round Indicate the number of pulses required for the stepper motor to complete one revolution.
Focal length： the distance from the inner sensor to the engraving surface, the laser head will go up or down according to this focal length number when pressing “Auto” button the focal length number can be input by press the button “Send focus”
Real focal length: display the current focal distance while the laser head goes up or down
Send focus: Input the number displayed on “Real Focal length” onto “Focus Length”
OK: parameters will be stored by the control board by clicking “OK”. The Auto button will run according the parameters
EN/中： change the language into Chinese or English


Lift up: the laser head will continuously go up when pressing the button when tapping the button



Auto focus: the laser head will move the right focal length according to the setting parameters above



Stop: the laser head will stop when it is continuously moving

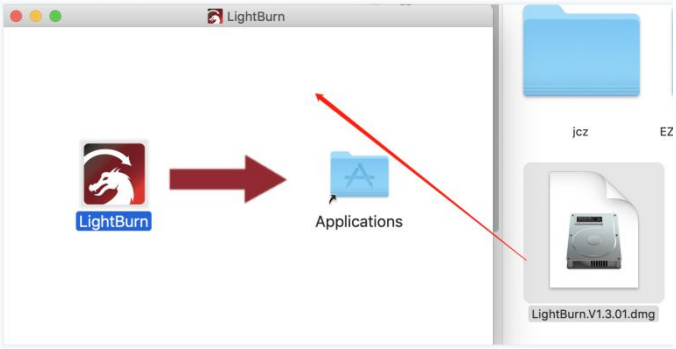
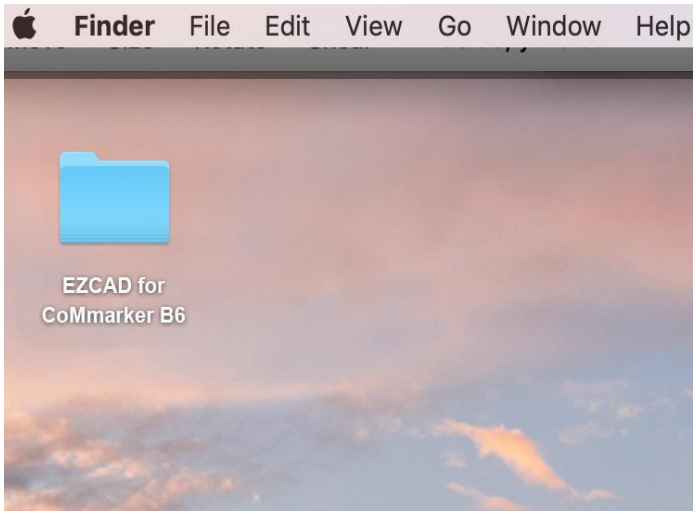
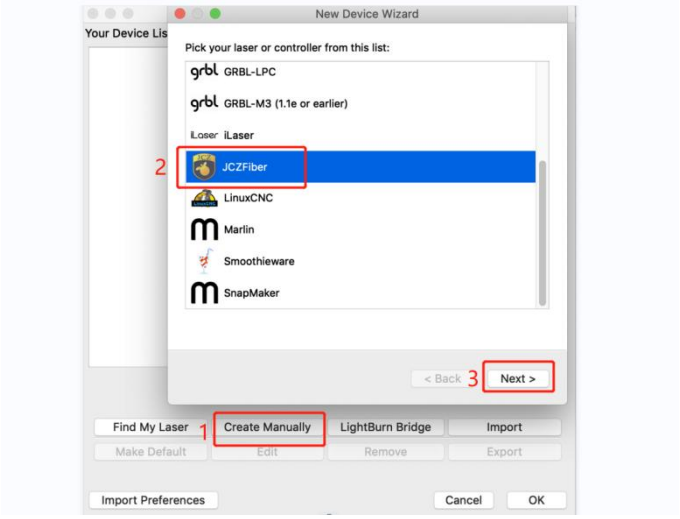
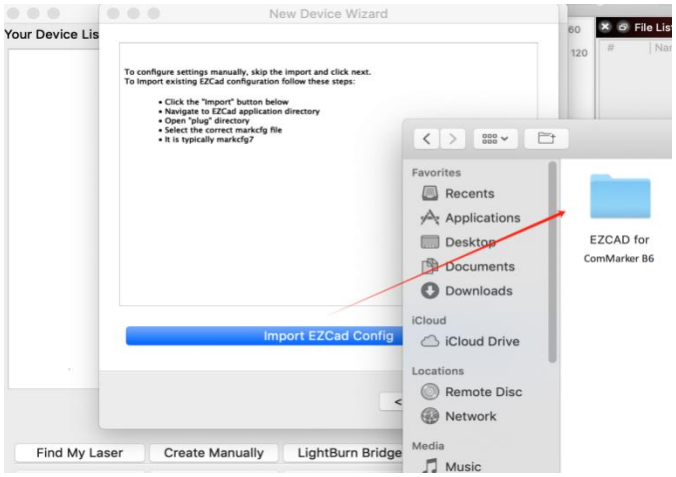
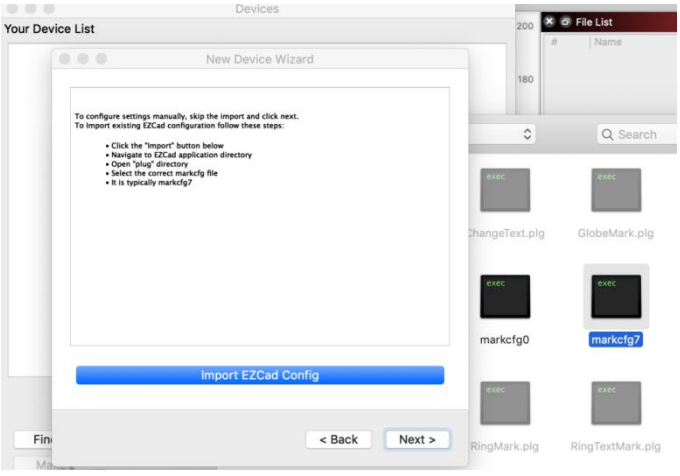

Lift down: the laser head will continuously go down when pressing the button when tapping the button

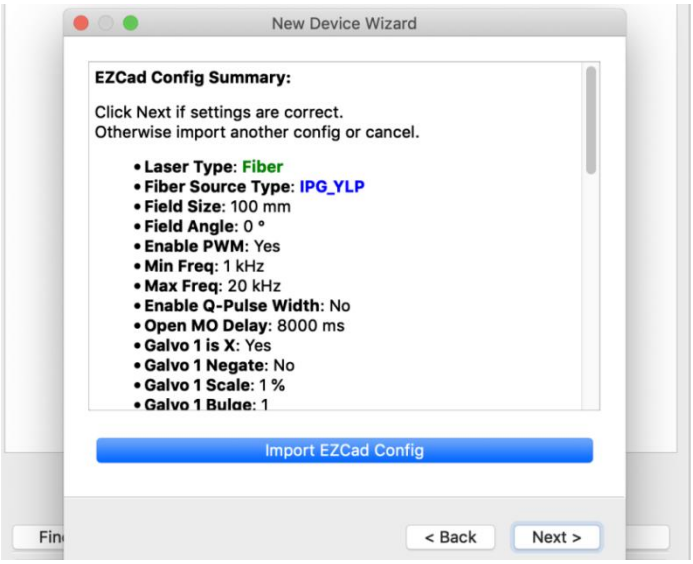
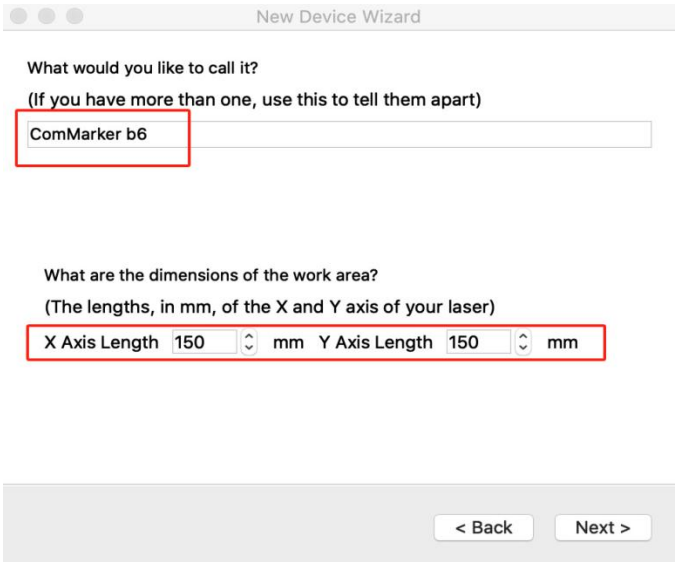
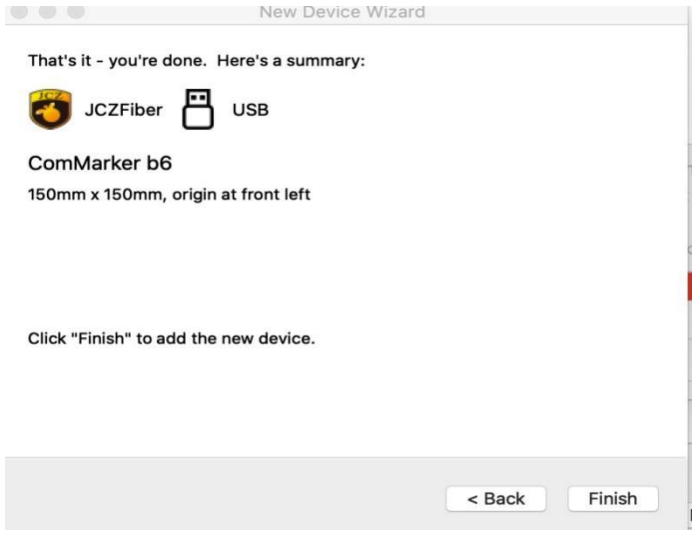
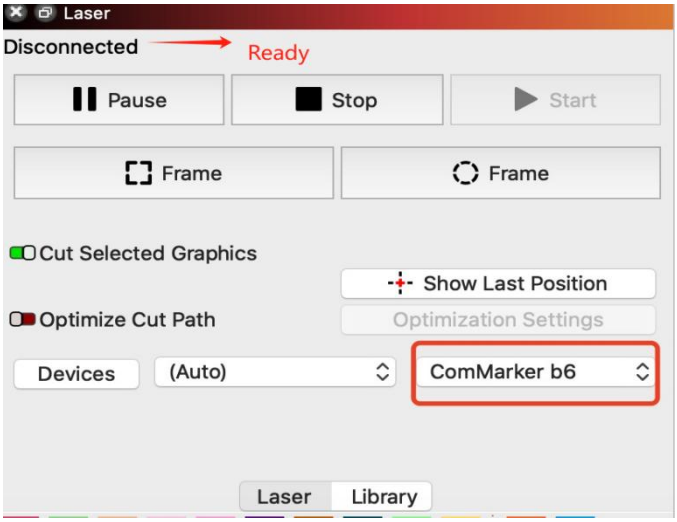
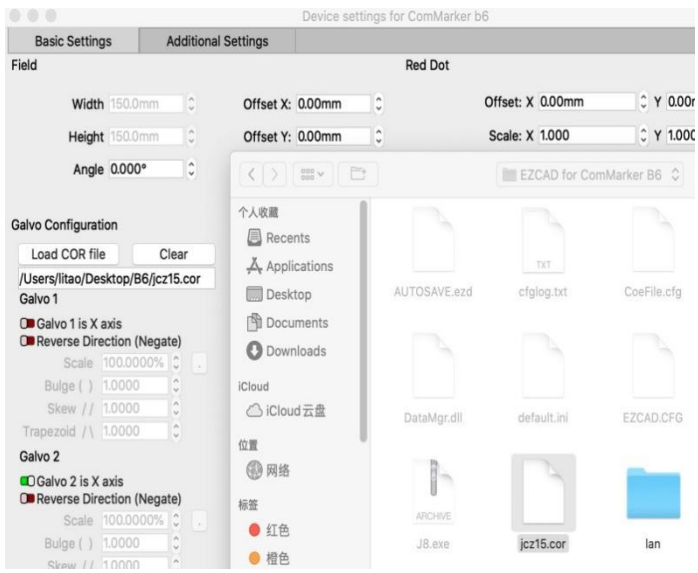
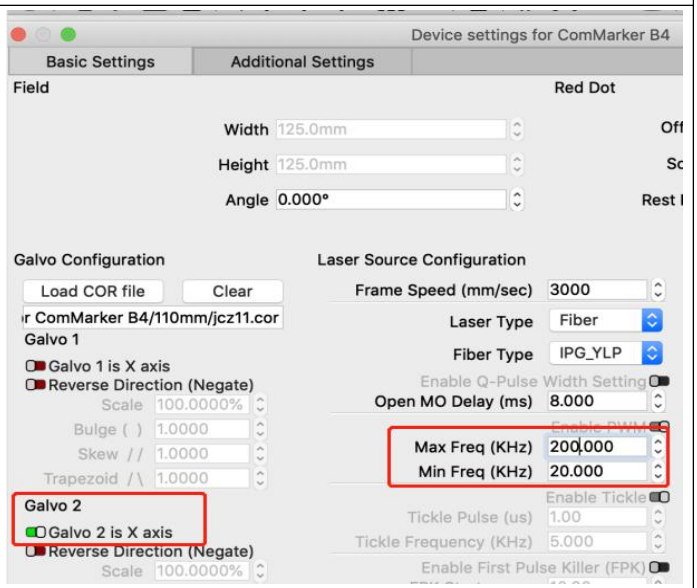
Lightburn Operation Tutorial (For MacOS)

1. Download and install Lightburn software from the website

Note: It is not compatible with our machine if your Lightburn version is lower than V1.3.01, please download a new version and reinstall.

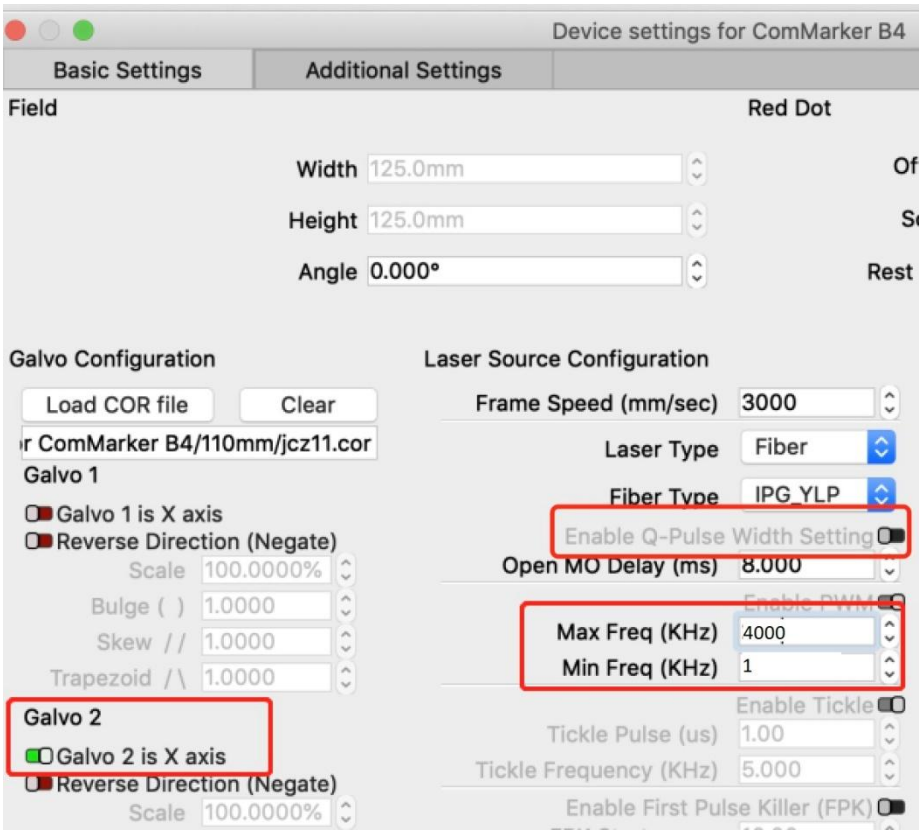
And Lightburn is a paid software with 1 month free trial, you need to buy the licence key for Pro version.

<table>
<tr><td>1. Down load the software from the website: https://lightburnsoftware.com/pages/trial-version-try-before-you-buy</td><td>2.Find downloaded file Lightburn V1.3.01, and drag Lightburn icon to the Applications folder</td></tr>
<tr><td> 
</td><td> 
</td></tr>
<tr><td>3.Copy the folder “EZCAD for ComMarker B6” from the USB to the desktop</td><td>4. A device window will pop out if the software is open the first time, click “Create Manually”, Select “JCZFiber”
Note: Make sure the machine is on and connected via USB</td></tr>
<tr><td> 
</td><td> 
</td></tr>
<tr><td>5.Import EZCAD Config and select the folder</td><td>6.Select folder “EZCAD for ComMarker B6”---->“ Plug”, and choose “markcfg7” and click “next”</td></tr>
<tr><td> 
</td><td> 
</td></tr>
</table>

7.You can see the following window if everything goes right, and click “next”	8. Change the name if necessary, and input the right X and Y Axis Length according to the lens. Click “next”
	
9.Click “finish” and it will in the device list	10. The status will change into “Ready” when the machine is on and connected via USB
	
11. Go to “Basic Settings” from “Device setting” button, click “Load COR file”, Select “EZCAD for ComMarker B6”, Choose file “JCZ15.cor”, Click “OK”	12. Change the setting of Freq “Min 20”, “Max 200”. Switch on “Galvo 2”. And it is ready to control the machine with Lightburn.***(Please refer to step 13 for MOPA machine)***
	

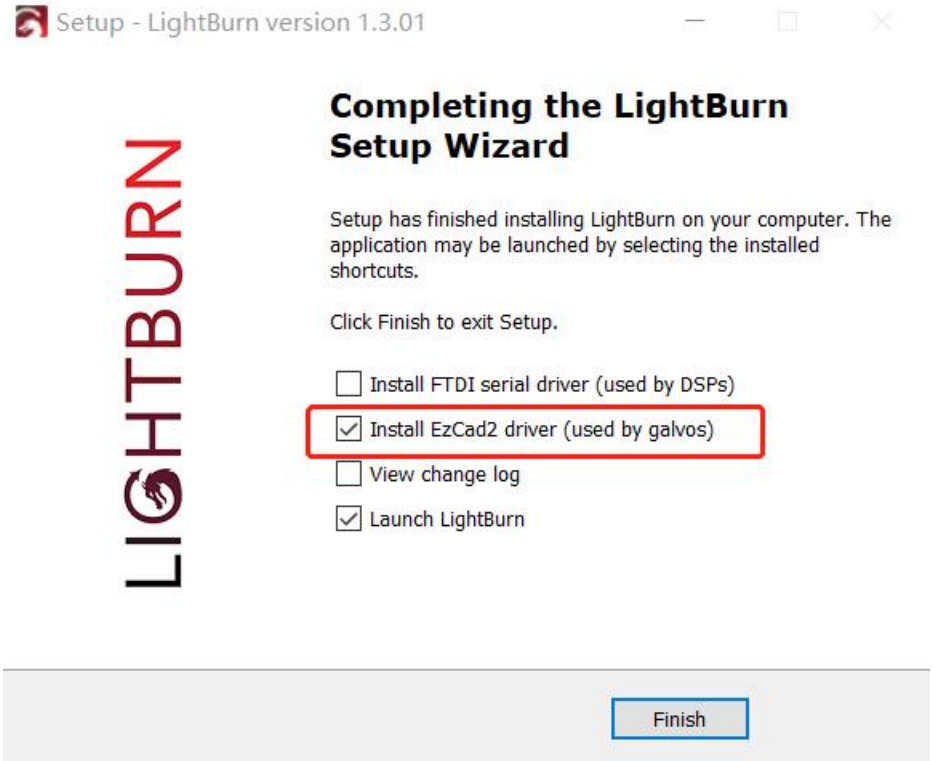
Special Setting for JPT MOPA fiber

13. Change the setting of Freq “Min 1”, “Max 4000”. Switch on “Galvo 2”. Switch on the button “Enable Q-Pulse Width Setting” (Make sure the Fiber Type is “IPG _YLP” or “JPT”)



Lightburn Operation Tutorial (For Window system)

Refer to the operation on MacOS, please don’t forget to install the driver when installing Lightburn software as the following picture shows:



Maintenance

1. Maintenance Overview



Unless otherwise specified, perform adjustment and maintenance of this device only when the power is turned off and the power supply has been disconnected. Only trained and skilled professionals should modify or disassemble this device.

2. Regular Maintenance Procedures

lKeep the workroom clean and dust-free at all times.

lEnsure the device is fully powered off when not in use.

lCover the galvanometric lens when it is not in use.

lClean the worktable after use with a cloth wetted with more than 75% rubbing alcohol.

l**Never** clean this device with abrasive or caustic cleansers, aerosol sprays, or enough water to enter any electrical component. Always allow surfaces to fully dry before further use.

lWhen removing dust from the device's vents using a vacuum, **ONLY** use the lowest power setting to avoid damage to internal components.



No other servicing should be done by the operator. Do not attempt to service or replace other parts yourself.

3. Troubleshooting Guidance

Potential Problems	**Possible Solutions**
No Laser Output	Correct the focus by adjusting the height of the laser arm.
	Correct the software parameters if they are invalid or mistaken.
	Have a technician fix or establish the connection between the laser and the motherboard.
	Have a technician fix or establish the connection between the laser and its power supply.
	If either the fiber laser source or its power supply has worn out, have a technician replace them.
No Engraving despite Laser Output	Confirm that the material can be safely engraved with this device.
	Correct the focus by adjusting the height of the laser arm.
	Adjust the software parameters to create greater intensity.
	Have a technician check the control panel, scanning lens, and their power supply. Correct any problems or replace the parts.
Other Laser Errors	Have a technician check the fiber. Have a technician check the fiber laser source and the mainboard. Correct any problems or replace the parts.

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Customer Service:

- For more video and supportive materials, please visit our official website at: *www.commarker.com*

- For Laser Engraver technical support and service please email: *support@commarker.com*